A counterexample to rising-factorial Toeplitz stability

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Task 1. Resolve the Hurwitz-stability conjecture

For $(x)_k = x(x+1)\cdots(x+k-1)$, $(x)_0 = 1$, Karp [1] defines

$$Q_n^{\alpha,\beta}(x) = \sum_{k=0}^{n} f_k f_{n-k} \binom{n}{k} \big[(x+\alpha)_k (x+\beta)_{n-k} - (x+\alpha+\beta)_k (x)_{n-k}\big].$$

Conjecture 2 asserts that, for $\alpha, \beta > 0$ and $n \geq 3$, the inequalities $f_k^2 > f_{k-1} f_{k+1}$, $1 \leq k < n$, imply that every zero of $Q_n^{\alpha,\beta}$ has negative real part.

Proposition 1. *The conjecture is false. Take $n = 12$, $\alpha = 1$, $\beta = 40$ and*

$$(f_0, \ldots, f_{12}) = (1, 45, 46, 47, 48, 49, 50, 49, 48, 47, 46, 45, 1).$$

Then $Q_{12}^{1,40}$ has exactly one zero in

$$|z - (1 + 22i)| < \frac{1}{4}.$$

In particular that zero has real part greater than $3/4$.

Proof. All entries are positive. The eleven strict log-concavity differences are

$$(f_k^2 - f_{k-1} f_{k+1})_{k=1}^{11} = (1979, 1, 1, 1, 1, 99, 1, 1, 1, 1, 1979).$$

The hypotheses are therefore satisfied, including strictness at every required index.

Direct polynomial multiplication gives

$$Q_{12}^{1,40}(x) = 5280 P(x), \qquad P(x) = \sum_{j=0}^{10} p_j x^j, \tag{1}$$

with the integer coefficients in the table below. We supply an exact Rouché certificate for the stated disk. Let $c = 1 + 22i$ and write

$$P(c+w) = \sum_{k=0}^{10} d_k w^k, \qquad d_k = \sum_{j=k}^{10} \binom{j}{k} p_j c^{j-k} \in \mathbb{Z}[i].$$

Define

$$U_k = 1 + \left\lfloor \sqrt{(\Re d_k)^2 + (\Im d_k)^2} \right\rfloor, \qquad L = \left\lfloor \sqrt{(\Re d_1)^2 + (\Im d_1)^2} \right\rfloor.$$

Thus $|d_k| < U_k$ and $L \leq |d_1|$. All the quantities involved are integers, and each floor is checked by squaring adjacent integers.

k	p_k	U_k
0	8850973475178928800	816065210715919818
1	2242108441860414480	4713796926694788092
2	269652359703530916	1088281691451296882
3	20855984967227000	110502641723796844
4	1180285415416015	6548680002689171
5	51443738871420	250999762004696
6	1706782696773	6478398028259
7	40875054000	112919703186
8	653614185	1283564680
9	6181900	8631961
10	26111	26112

For example, the first two Taylor coefficients are

$$d_0 = 189742720030323200 - 793700276134708800i,$$
$$d_1 = 4623392455279316640 + 918762031531300000i,$$

and

$$L = 4713796926694788091 > 0.$$

With $r = 1/4$, the exact rational calculation from the table is

$$\frac{U_0 + \sum_{k=2}^{10} U_k r^k}{Lr} = \frac{232216395171843963627545}{308923395387869632331776} < \frac{4}{5} < 1. \tag{2}$$

On $|w| = r$, this proves

$$|P(c+w) - d_1 w| \le |d_0| + \sum_{k=2}^{10} |d_k| r^k < Lr \le |d_1 w|.$$

Rouché's theorem implies that $P(c+w)$ and $d_1 w$ have the same number of zeros, with multiplicity, in $|w| < r$, namely one. The disk is contained in $\{\Re z > 3/4\}$. Equation (1) now gives the desired contradiction. □

Certificate reproducibility. The accompanying standard-library Python verifier constructs (1) by integer convolution and independently checks the identity at thirteen distinct integers (enough for degree at most twelve). It computes the Gaussian integer d_k in two independent ways: the binomial formula above and Horner substitution. Integer square roots then verify every table entry and the rational bound (2). No floating-point zero location is used in the proof.

Scope. The example has a positive input sequence with no internal zeros, and all eleven coefficients of P are positive. It refutes the universal Hurwitz-stability assertion, while leaving the separate coefficient-positivity assertion untouched. Complex conjugation also gives a zero in the reflected disk centered at $1 - 22i$. No claim of minimal degree is made.

Task 2. Find the first possible failure

Determine the smallest n for which the conjecture fails, and prove stability for all smaller degrees or exhibit a smaller counterexample.

Task 3. Classify the stable parameter range

Determine the pairs (α, β) for which $Q_n^{\alpha,\beta}$ is Hurwitz stable for every strictly log-concave positive input and every $n \geq 3$, including the special case $\alpha = \beta = 1$.

Task 4. Settle coefficient positivity

Determine whether strict log-concavity always forces every coefficient of $Q_n^{\alpha,\beta}$ to be positive, and identify the exact hypotheses needed if this weaker assertion fails.

Task 5. Strengthen the input hypotheses

Find verifiable stronger conditions on the input sequence that restore Hurwitz stability, and determine how those conditions depend on n, α and β.

References

[1] D. B. Karp, *Positivity of Toeplitz determinants formed by rising factorial series and properties of related polynomials*, Zap. Nauchn. Sem. POMI **404** (2012), 184–198, Conjecture 2, p. 185; translated in J. Math. Sci. **193** (2013), 106–114. https://www.mathnet.ru/eng/znsl5267. Preprint: https://arxiv.org/abs/1203.1482.